

MCS- 215

Security and Cyber Laws

Q-1.(a) what are the three pillars of digital security? what is the need for digital security?

Ans- The three pillars of digital security, often referred to as the CIA triad, are.

1. ** Confidentiality : **

Ensuring that information is accessible only to those who have the proper authorization. This involves measures such as encryption, access controls.

(b) Explain the following in the context of security issues / attacks:

(i) DDos attacks

(ii) Malware

(iii) Crypto-jacking

Ans- let's delve into the explanations of DDos attacks, malware and crypto-jacking in the context of security issues / attacks!

(i) ** DDos Attacks (Distributed Denial of Service)! **

DDos attacks involve overwhelming a target's online service or network by flooding it with traffic from multiple sources, making it difficult or impossible for legitimate users to access the service. The "distributed" aspect comes from the use of network.

(c) Explain the term Cyber Security intrusion detection with the help of an example.

Ans- ** Cyber Security Intrusion Detection: **

Intrusion detection in cybersecurity refers to the process of monitoring and analyzing network or system activities for signs of unauthorized access, attacks or security breaches.

1. ** Baseline Establishment: **

The intrusion detection system initially establishes a baseline of normal network behaviour, this includes typical patterns of data transfer, user access, and system activities during regular operation.

2. ** Anomaly Detection: **

As the IDS continues monitoring, it detects an anomaly.

(d) what are the laws related to unauthorized access and web jacking?

⇒) laws related to unauthorized access and web jacking can vary significantly across jurisdictions. In many cases, these offenses are covered under broader cybercrime or computer crime legislation. Here are some general principles and examples, but it's important to note that the specifics can differ based on the country or region.

1. ** Computer Fraud and Abuse Act (CFAA) - United States: **

- The CFAA is a comprehensive U.S. federal law that addresses unauthorized access to computer systems.

2. ** Computer Misuse Act - United Kingdom: **

- The Computer Misuse Act criminalizes unauthorized access to computers.

Q-2 Explain the following terms with the help of an example of each.

(a) Function-based substitution cipher.

⇒ ** Function-Based Substitution cipher

A function-based substitution cipher involves the use of a mathematical function to replace or substitute characters in the plaintext with corresponding characters in the ciphertext. This type of cipher relies on a specific algorithm or mathematical to perform the substitution, making it more complex than traditional ciphers.

** Examples: **

Let's consider a simple function-based substitution cipher where the mathematical function is to shift each letter.

(b) Steganography

=> Steganography is the practice of concealing or hiding information within other non-secret data, making it difficult for an observer to detect the existence of the hidden information.

Key aspects of steganography include:

1. ** Carrier Medium **

- The carrier medium is the object or file in which the secret information is hidden. This can be an image, audio file, video, text, or any other digital or physical medium.

2. ** Hidden Message **

- The hidden message, also known as the payload or steganogram, is the information that is concealed within the carrier.

(c) RSA algorithm

=> The RSA algorithm is a widely used public-key cryptosystem that facilitates secure data transmission and digital structures.

The acronym RSA stands for the surnames of its inventors, Ron Rivest, Adi Shamir, and Leonard Adleman. It was introduced in 1977 and remains a fundamental component of modern cryptography.

Here is an overview of how the RSA algorithm works:

1. ** Key Generation: **

- RSA involves the generation of a pair of public and private keys.

- The public key is used for encryption, and it can be freely distributed.

(d) Hash functions

Ams- A hash function is a mathematical function that takes an input (or "messages") and produces a fixed-size string of characters, which is typically a hexadecimal number.

The output, often referred to as the hash value or hash code, is unique to the input data.

Here are some key characteristics and uses of hash functions:

1. ** Deterministic **

A hash function is deterministic, meaning that for a given input, it will always produce the same hash value. This property is crucial for consistency in various applications.

2. ** Fixed output length: **

3. (a) list practices for implementing the CIA triad in data security.

Ams-

Implementing the CIA triad (Confidentiality, Integrity, and Availability) in data security involves adopting various practices and measures to safeguard information. Here are key practices for each component of the CIA triad.

1. ** Encryption **

Use strong encryption algorithms to protect sensitive data, both in transit and at rest. Encrypting data ensures data that even if unauthorised access occurs, the information remains unreadable without the appropriate decryption key.

* ** Access Controls: **

=> Implement access controls to restrict access to sensitive data based.

(b) Explain the following:

(i) Phishing attacks

(ii) Ransomware attacks

(iii) state-sponsored attacks

Ans (i) **Phishing Attacks:**

Definition:-

Phishing is a type of cyber attack where attackers use deceptive emails, messages, or websites to trick individuals into divulging sensitive information, such as involve social usernames, passwords, credit card number or other personal details.

Phishing attacks often involve social engineering tactics to exploit human psychology and manipulate recipients into taking actions that benefits the attacker.

(C) Explain the six principles of security management.

⇒ Security management involves the implementation and maintenance of measures to protect assets, information and systems from various threats.

The six principles of security management provide a framework for developing and maintaining effective security practices within an organisation.

γ ** Principle of Defense in Depth **:

Definitions:-

Defense in depth involves implementing multiple layers of security measures to provide redundancy and increase the overall resilience of a security system.

(d) Explain the terms -

(i) Security audit

(ii) Security and usability

=) ** Definitions: **

A security audit is a systematic evaluation of an organisation's information systems, processes, and policies to assess the effectiveness of its security controls.

The goal is to identify vulnerabilities, measure compliance with security policies and standards, and recommend improvements to enhance overall security.

** Key components of a Security Audit: **

1. ** Risk Assessment: **

evaluate potential risks and threats

Q-4(a) why is there a need to regulate cyberspace? Explain, giving reasons.

=> Regulating cyberspace is essential due to the increasing dependence on digital technologies and the internet in various aspects of daily life, business, and critical infrastructure.

The need for regulation arises from several reasons:

1. ** Security Concerns: **

- ** Cyber Threats: **

cyberspace is susceptible to a wide range of cyber threats, including hacking, malware, phishing, and ransomware attacks.

Regulation helps establish security standards and measures to protect individuals, organisations, and government from these threats.

(b) Explain the role of filtering devices and rating scales in regulating internet content.

=) Filtering devices and rating scales play significant roles in regulating internet content, contributing to the protection of users, especially children, from inappropriate or harmful material.

These tools are used to control access based on content based on its nature, ensuring a safer and more controlled online environment.

Here's an overview of their roles:

1) ** Filtering Devices: **

** Definition: **

Filtering devices, often implemented as software or hardware solutions, are designed to block

(c) what is the UNCITRAL model law?
Explain its doctrines and parts.

⇒ The UNCITRAL Model law, developed by United Nations Commission on International Trade law (UNCITRAL), stands for the United Nations Commission on International Trade law Model law on Electronic commerce.

It is a set of guidelines and recommendations aimed at providing a legal framework for the use of electronic communications and transaction in international trade.

The Model law was first adopted in 1996 and has since been widely influential in shaping electronic commerce legislation globally.

** Key Doctrines of the UNCITRAL Model law: **

1. ** Technology Neutrality: **

(d) what are the international initiatives for the regulation of cyberspace?

⇒ Several international initiatives aim to establish norms, standards, and cooperation frameworks for the regulation of cyberspace. These initiatives involve collaboration among countries, international organisations and other stakeholders to address the challenges and threats posed by activities in the digital realm.

Some notable international initiatives include:

1. ** United Nations Group of Government Experts (UNGGE): **

** Purpose : **

The UNGGE is a forum of United Nations member states to discuss and recommendations.

Q-3(a) Explain the classification of cyber-crimes with the help of examples

⇒) cybercrimes encompass a broad range of illegal activities facilitated by computer networks, the internet, and digital technologies.

classifying cybercrimes helps in understanding and addressing specific types of offenses in the digital realm.

Here are common classifications of cybercrimes along with examples:

1. ** Cyber Fraud: **

** Definition: **

cyber fraud involve deceptive practices to gain unauthorized access to sensitive information, financial resources, or intellectual property.

(b) How is Computer Contaminant defined under Section 43 of the Information Technology Act 2000?

⇒ Section 43 of the Information Technology (IT) Act, 2000 in India defines the terms "computer contaminant",

This section outlines offenses related to unauthorized access, damage, or disruption to computer system, and it includes provisions related to computer contaminants.

Here is the definition of "computer contaminants" per section 43:

** Section 43: Unauthorized Access to Computer systems, Data Alteration, and introduction of computer contaminants: **

(i) Unauthorized Access: **

(c) list any six offences, as per the introduction of information technology Act, 2000:

=> The information Technology (IT) Act, 2000 in India addresses various offences related to the misuse of technology and digital resources.

Here are six offences outlined in the IT Act:

1. ** Unauthorized Access and Hacking (Section 43): **

- ** offence: **

unauthorized access to computer systems, networks, or data without the owner's permission.

- ** Penalty: **

compensation for damages and a fine.

(c) list any six offenses, as per the introduction of information technology Act, 2000:

=> The information Technology (IT) Act, 2000 in India addresses various offenses related to the mis-use of technology and digital resources.

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(a) what are the liabilities of network service providers? explain.

=> network service providers play a crucial role in facilitating communication and connectivity in the digital world.

They also bear certain liabilities and responsibilities as intermediaries in the transmission of information.

The liabilities of network service providers are typically outlined in legal frameworks to ensure responsible and ethical use of their service. These liabilities can vary across jurisdictions, but common aspects include:

1. ** Duty to Exercise Due Diligence (Section 79 of the Information Technology Act, 2000 - India)**

Q 6(a) Explain the following forms of IPR and related regulatory framework:

(i) copyrights and related rights

(ii) Patents

(iii) Trademarks.

Ans- ** Intellectual Property Rights (IPR) and Regulatory Framework **

** (i) Copyrights and Related Rights: **

** Definitions: **

Copyrights protect original literary, artistic, and musical works.

Related rights extends protection to performers, producers of sound recordings, and broadcasting organisations.

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** Definition: **

Copyrights protect original literary, artistic, and musical works. Related rights extend protection to performers, producers of sound recordings, and broadcasting organizations.

** Regulatory Framework: **

- ** Copyright Act, 1957 (India): **

The copyright Act in India governs the protection of literary, artistic, and musical works.

It grants exclusive rights to creators and copyright owners to reproduce, distribute, and display their works.

- ** Berne Convention: **

An international treaty that sets standards for copyright protection. Many countries, including India, are conventions.

(b) what is meant by the terms- linking
1. in-linking and framing in the
context of IPR?

⇒ In the context of intellectual Property Rights (IPR) and the digital environment, the digital environment, the terms "linking", "in-linking" and "framing" refers to different ways in which content from one source may be integrated or connected with another.

These actions can raise legal and copyright issues, and their implications may vary based on regional laws and regulations.

Here's an explanation of each other:

1. ** linking: **

** Definition: **

linking involves providing a hyperlink from one web page to another.

(c) what are domain disputes? explain with the help of an example.

=> Domain name disputes typically arise when there is a conflict over the ownership, use, or registration of a specific domain name.

These disputes often involve claims of trademark infringement, cybersquatting, or other violations of intellectual property rights.

The resolution of domain name disputes is commonly handled through mechanisms such as the Uniform Domain - Name Dispute Resolution Policy (UDRP) or legal proceedings.

The dispute resolution process could unfold in the following ways:

1. ** UDRP complaint; **

- ABC Electronics can file a complaint under the Uniform Domain Name Dispute - Resolution Policy (UDRP).

UDRP is a process established by the Internet Corporation for Assigned Names and Numbers (ICANN) to resolve domain name disputes.

2 ** Complaint Circumstances! **

- ABC Electronic would need to demonstrate three essential elements for a successful UDRP complaint.
- The domain name is identical or confusingly similar to a trademark or service mark in which ABC Electronics has rights.
- The registrant has no legitimate rights or interests in the domain name.